

MATH 121: COMMUTATIVE ALGEBRA

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1. RINGS

Important Example: Let X be a “geometric set”, and consider the ring

$$\text{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \text{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \text{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\text{Hom}(X, \mathbb{R})^\times = \text{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. Let $a \in R, a \neq 0$

- a is a unit if and only if $\exists b \in R$ such that $ab = 1$,
- a is a zero divisor if and only if $\exists b \in R \setminus \{0\}$ such that $ab = 0$,

Lemma 1.2. The set of units $R^\times = \{a \in R \mid a \text{ is a unit}\}$ is a group under multiplication.

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma). If $\phi : R \rightarrow S$ is a ring homomorphism, then it follows from the definition that

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. $\mathbb{Z}$ is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring R ,

$$\text{Hom}(\mathbb{Z}, R) = \{1 \rightarrow 1_R\}$$

We don't know anything about homomorphisms to the initial object.

2. IDEALS

Lemma 2.1. If $\phi : R \rightarrow S$ is any ring homomorphism and $J \subset S$ is an ideal $\phi^{-1}(J)$ is also an ideal.

Proof: Worksheet Q4.1.

Theorem 2.2. If $I \subset R$ is an ideal then,

- (1) There exists a ring R/I and a surjective ring homomorphism $\pi : R \rightarrow R/I$ such that $\ker(\pi) = I$. Moreover R/I and π are unique upto isomorphism.
- (2) There is an inclusion preserving bijection

$$\{\text{Ideals in } R/I\} \leftrightarrow \{\text{Ideals in } R \text{ containing } I\}$$

$$\bar{J} \leftrightarrow \pi^{-1}(\bar{J})$$

Proof.

- (1) Construct R/I explicitly.
- (2) I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \rightarrow R/I = R/\sim$$

$$a \mapsto \bar{a} = a + I$$

Let $\bar{J}$ be an ideal in R/I , and J be an ideal in R and define the maps

$$\phi : \bar{J} \rightarrow \pi^{-1}(\bar{J}), \quad \psi : J \rightarrow \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = \text{id}$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\bar{J})$. Note that

$$\psi \circ \phi(\bar{J}) = \pi(\pi^{-1}(\bar{J})) = \bar{J}$$

since π is surjective. Thus $\psi \circ \phi = \text{id}$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\bar{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\bar{J})$ looks like $x + I$, and $\pi^{-1}(\bar{J}) = J + I$. Moreover since $\bar{0} \in \bar{J}$, we have $\pi^{-1}(\bar{0}) \subseteq \pi^{-1}(\bar{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = \text{id}$.

□

Exercise: Describe the ideals in

- (1) $\mathbb{Z}/\langle 24 \rangle = \{\langle 0 \rangle, \langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle\}$
- (2) $\mathbb{Z}/\langle n \rangle = \{\langle d \rangle : d|n\}$
- (3) $\mathbb{Z}/\langle 3 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (4) $\mathbb{Z}/\langle 5 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (5) $\mathbb{C}[x, y]/\langle y - x^2 \rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
- (6) $\mathbb{C}[x]/\langle x^n \rangle = \{\langle x^i \rangle : 0 \leq i < n\}$

Lemma 2.3. If $\phi : R \rightarrow S$ is a ring homomorphism with $\ker(\phi) \supseteq I$ then there exists a unique morphism $\tilde{\phi} : R/I \rightarrow S$

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ \phi \downarrow & \swarrow \tilde{\phi} & \\ S & & \end{array}$$

Universal Property of Quotients: Upto isomorphism R/I is the ring such that

$$\text{Hom}(R/I, S) = \{\phi : R \rightarrow S : I \subset \ker(\phi)\}$$

We want to define $\tilde{\phi}(\bar{a}) = \tilde{\phi}(a + I) = \phi(a)$ and $\tilde{\phi}(\bar{0}) = 0$.

Lemmat 2.4. Maximal Ideal $\implies$ Prime Ideals.

Definition 2.5. The spectrum of a ring R is the set

$$\text{Spec}(R) = \{\text{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$\mathfrak{m}\text{Spec}(R) = \{\text{Maximal Ideals in } R\}$$

$\text{Spec}(R)$ is where the ‘geometry lives’. Note that $\mathfrak{m}\text{Spec}(R) \subset \text{Spec}(R)$.

Example: $R = \mathbb{Z}[\sqrt{-5}]$. 2 is irreducible but not prime. $R/\langle 2 \rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example: $R = \mathbb{C}[x, y]/\langle x^3 - y^2 \rangle$. x is irreducible, but $\langle x \rangle$ is not prime. $R/\langle x \rangle = \mathbb{C}[y]/\langle y \rangle$ and y is a zero divisor.

Theorem 2.6. Let R be a non-zero ring.

- (1) R has a maximal ideal.
- (2) If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.

Proof. **Zorn’s Lemma:** Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \dots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i , then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{\text{Proper ideals in } R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

- (1)
 - $P \neq \emptyset$ since $\langle 0 \rangle \in P$.
 - Given a chain in P

$$I_1 \subseteq I_2 \subseteq \dots$$

Then, define $B = \bigcup I_j$. Note that B is an ideals since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j , and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.

- (2) Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $\mathfrak{m}\text{Spec}(R) \neq \emptyset$ and $\text{Spec}(R) \neq \emptyset$. □

3. ZARISKI TOPOLOGY

Definition 3.1. If $S \subseteq R$ is any subset of R ,

$$\mathbb{V}(S) = \{[P] \in \text{Spec}(R) | S \subseteq P\} \subseteq \text{Spec}(R)$$

Definition 3.2. An algebraic subset of $\mathbb{K}^n$ of some field $\mathbb{K}$ is a set of the form

$$\mathbb{V}(S) = \{P \in \mathbb{K}^n | f(P) = 0, \forall f \in S\}$$

for a subset $S \subset \mathbb{K}[x_1, \dots, x_n]$

Examples:

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^2$

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^3$

$$\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 - y) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x^2 = y\}$$

- $\mathbb{V}(x^2 - xy) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{R}[x]$

$$\mathbb{V}(x^2 + 1) = \emptyset$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{C}[x]$

$$\mathbb{V}(x^2 + 1) = \{i, -i\}$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_3[x]$

$$\mathbb{V}(x^2 + 4) = \emptyset$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_p[x]$

TODO:

- $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subseteq \mathbb{R}[x, y]$

$$\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) = \mathbb{V}(x^3(x - y)^3) = \{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(n(x^2 + y^2 + z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$

TODO:

- $\mathbb{V}(\text{a collection of any linear polynomials in } \mathbb{K}[x_1, \dots, x_n])$

TODO:

Definition 3.3. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is

$$\mathbb{I}(V) = \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(p) = 0, \forall p \in V\}$$

Definition 3.4. If $\mathfrak{M} \subseteq R$ is a maximial ideal, the residue field of $\mathfrak{M}$ is $R/\mathfrak{M}$.

Definition 3.5. Let $\mathbb{K}$ be a field. A $\mathbb{K}$ algebra is the data of a ring R together with a fixed ring homomorphism

$$\alpha : \mathbb{K} \rightarrow R$$

Definition 3.6. If $(R, \mathbb{K} \xrightarrow{\alpha} R)$ and $(S, \mathbb{K} \xrightarrow{\beta} S)$ are $\mathbb{K}$ -algebras, a ring map $\phi : R \rightarrow S$ is a morphism of $\mathbb{K}$ -algebras iff

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\phi} & \mathbb{S} \\ \alpha \uparrow & \nearrow \beta & \\ \mathbb{K} & & \end{array}$$

Lemmact 3.7. If $\mathbb{R}$ is $\mathbb{k}$ -algebra, $\mathbb{R}/I$ has a natural $\mathbb{k}$ -algebra structure compatible with the quotient map.

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\pi} & \mathbb{R}/I \\ \alpha \uparrow & \nearrow \exists! & \\ \mathbb{k} & & \end{array}$$

Definition 3.8. A valuation ring of a field $\mathbb{K}$ is a subring $\mathbb{R} \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in \mathbb{R}$ or $x^{-1} \in \mathbb{R}$.

4. GRÖBNER BASIS

Remark 4.1. *Ideal Membership Problem:* Suppose $I \subseteq \mathbb{R}[x_1, \dots, x_n]$ an ideal. How can we algorithmically determine if $f \in I$?

Definition 4.2. A ring $\mathbb{R}$ is Noetherian if and only if every ascending chain of ideals stabilizes (Ascending Chain Condition), i.e.

$$I_1 \subseteq I_2 \subseteq \dots$$

there exists $N \in \mathbb{Z}$ such that $I_k = I_{k+1}$ for all $k \geq N$.

Lemma 4.3. $\mathbb{R}$ is Noetherian if and only if every ideal is finitely generated.

Lemma 4.4 (Dickson's Lemma). Every monomial ideal in $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.

Theorem 4.5 (Hilbert's Basis Theorem). If $\mathbb{R}$ is a Noetherian ring, then $\mathbb{R}[x]$ is also Noetherian.

5. NULLSTELLENSATZ

Theorem 5.1 (Hilbert's Nullstellensatz I). If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.

Theorem 5.2 (Hilbert's Nullstellensatz II). If $\mathbb{K}$ is algebraically closed and $V = V(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $I(V) = \sqrt{\langle \mathcal{S} \rangle}$.

Theorem 5.3 (Hilbert's Weak Nullstellensatz). If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \dots, x_n]$ is an ideal then $V(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.

Theorem 5.4 (Hilbert's Nullstellensatz III). Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps I_V and V_V induce inverse, inclusion-reversing bijections:

$$\begin{array}{ccc} \left\{ \begin{array}{c} \text{algebraic subsets} \\ \text{in } V \end{array} \right\} & \begin{array}{c} \xrightarrow{I_V} \\ \xleftarrow{V_V} \end{array} & \left\{ \begin{array}{c} \text{radical ideals} \\ \text{in } \mathbb{K}[V] \end{array} \right\} \\ & 5 & \end{array}$$

Remark 5.5. $\mathbb{I}$ and $\mathbb{V}$ properties:

(1) If $\mathbf{f}\mathbf{g} \in \mathbb{I}(\mathbf{V})$ then $\mathbf{V} \subseteq \mathbb{V}(\mathbf{g}) \cap \mathbb{V}(\mathbf{f})$.